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Where Minds Pollinate

Class 9

Chemistry

Chapter 8 Energetics

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Unit # 8

Topic:- Energetics

Question no 1:- Define chemical energetics?

Answer:- Chemical energetics:-

"The Study of energy changes in chemical reaction is called chemical energetics"

Question no 2:- How energy changes occur in chemical reaction?

Answer:- Breaking bonds always consumes energy and binding always release energy.

1- If the energy released in forming a bond is greater than the energy expended in breaking the bond, there is a net release of chemical energy.

2- On the other hand, energy is absorbed when the energy expended in breaking a bond is greater than the energy released in forming the bond.

Question no 3:- Define System & Surrounding with eg?

Answer:- System:- The part of the universe that we want to focus our attention on is called System:

Surrounding:- The rest of the universe is called environment or Surrounding.

A System is usually a Substance that changes physically or chemically present within boundary.

For example:-

When studying the reaction of Limestone and hydrochloric acid solution in a test tube, Limestone and hydrochloric acid solution form a system. The test tube and everything around the test tube is the environment or surrounding.

Question no 4:- What are thermochemical reactions?

Answer:- "A balanced chemical equation which also shows heat change of a chemical reaction is called thermochemical equation. The branch of chemistry which deals with the heat or thermal energy changes in chemical reaction is called thermochemistry."

Example:-



$$\Delta H^\circ = -393.5 \text{ kJ}$$

Question no 5:- What are the 2 type of thermochemical reactions.

Answer:- There are two type of thermochemical reactions.

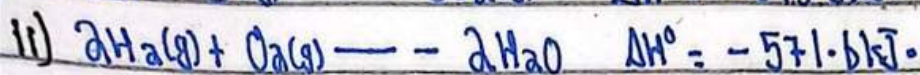
- 1- Exothermic reaction.
- 2- Endothermic reaction.

Question no 6:- Define Exothermic and Endothermic reactions.

Answer:- Exothermic Reaction:- "A chemical reaction that proceeds with the evolution of heat is called exo-

thermic reaction.

Example:-



• Endothermic Reaction:- "A chemical reaction that proceeds with the absorption of heat is called endothermic reaction."

Example:-



Question no 7:- Define enthalpy of reaction.

Answer:- Enthalpy of reaction:-

The amount of heat or thermal energy evolved in a chemical reaction is called enthalpy of reaction. Its sign is negative for exothermic and positive for endothermic reaction.

Enthalpy of reaction measured at 25°C (or 298 K) and one atmospheric pressure is known as standard enthalpy change. It is denoted by ΔH°

Example:-



Question no 8:- Define bond energy and bond dissociation energy.

Answer:- Bond energy and bond dissociation energy:-

When a chemical reaction occurs, old bonds are broken and new bonds are formed. Breaking bonds always require energy, and forming a bond always releases energy. The amount of energy required to break one mole of a particular bond to form neutral atoms is called the bond dissociation energy. In contrast, the amount of energy released when neutral atoms form one mole of a bond is called bond energy. The difference between the bond dissociation energy and the bond energy determines whether the reaction absorbs or releases energy.

Question no 9:- What is activation energy of chemical reaction?

Answer:- Activation energy:-

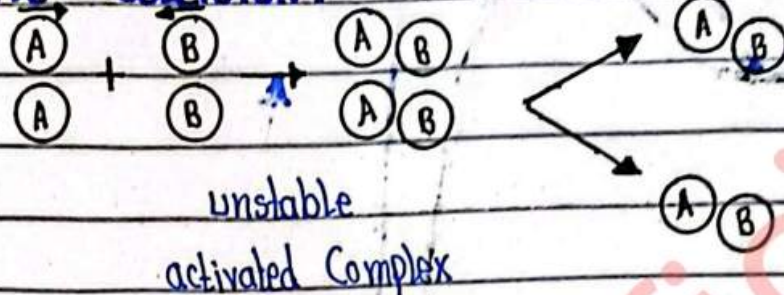
- The minimum amount of energy that, in addition to the average kinetic energy, particles must have for an effective collision is called the activation energy.
- For a chemical reaction to occur, the bonding atoms or molecules must collide with each other. These collisions can be effective or ineffective depending on the energy and the direction of the colliding particles. An effective collision can only occur if the energy of the colliding particles is high enough to overcome the repulsion between the electron clouds around the reacting particles.
- No reaction occurs if the energy of the reacting particles

is lower than the activation energy.

• The higher the activation energy, the lower the reaction rate.

For eg:- the reaction b/w A_2, B_2 molecules.

Effective collision:-



Ineffective Collision:-



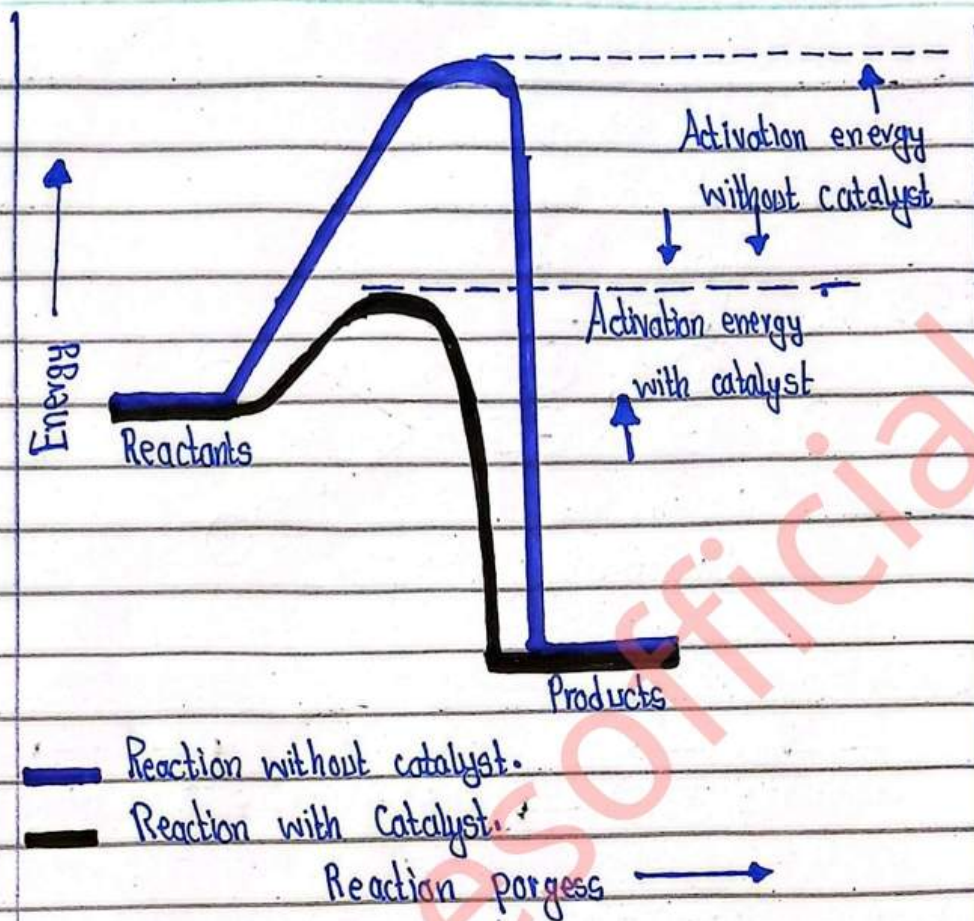
Question no 10:- Define Catalyst? What is its function in a chemical reaction?

Answer:- Catalyst:-

"A substance that accelerates a chemical reaction, but remains chemically unchanged at the end of the reaction, is called Catalyst." and the phenomenon is called Catalysis.

The Catalyst provides a new mechanism for the reaction with low activation energy. Thus, a Catalyst increases the rate of a reaction by lowering its activation energy. The Catalyst does not affect the overall thermodynamics or enthalpy of the reaction.

In the bodies of living organisms enzymes catalyst chemical reactions.



Question no 11:- Define Respiration?

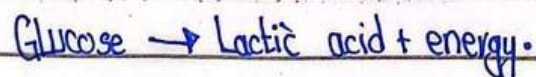
Answer:- Respiration:-

Respiration is a bio-chemical process in which energy is released from food in a biological system. During this process, glucose is oxidized in the body of living organisms and energy is released. Therefore, respiration is an exothermic reaction. There are two types of respiration process.

1. **Aerobic respiration:-** Respiration that requires oxygen to break down glucose to release energy is called aerobic respiration.



2. **Anaerobic respiration:-** Respiration that does not require oxygen to break down glucose to release energy is called anaerobic respiration.



Aerobic respiration release more energy than anaerobic respiration. Lipids are important substance for building our body. They also acts as reserve energy sources. Lipids can store very large amounts of energy in our body. When you exercise intensely, the oxidation of glucose is not enough for energy. At this stage, lipids are oxidized for energy.